Report on NNDL WiFi HAR and PI

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***Abstract*—Human Activity Recognition is a widely studied task that can be used in several applications, for example security systems and alarms. In this paper, we want to present two models based on Deep Learning techniques that can efficiently deal with Activity Recognition (AR) and Person Identification (PI) tasks. Our models can distinguish different activities performed by different persons, across different time-spans and environments. The recognition process is based on WiFi signals and Doppler effect because by exploiting the shift between traces we can tell if something is moving in an environment. To achieve this, we trained and tested our models on an already processed dataset of Doppler traces [1] [2]. The two models mainly differ in how they handle similarities between processed data by using supervised and unsupervised contrastive approaches [3], but they share some points in common because both have been developed starting from an already built framework [4]. Despite the differences, both models achieve very good results that can be summarized in an average accuracy of 96% on AR for both the models and 96% on PI for the unsupervised model, the supervised one focuses only on the activity recognition task.**

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***Index Terms*—Human Activity Recognition, Person Identification, Wi-Fi Sensing, Self-Supervised Learning, Contrastive Learning.**

I. Introduction

II. Related Work

III. Processing Pipeline

IV. Results

V. Concluding Remarks

References

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